Clinical Emergency Protocol: Acute Ischemic Stroke Management

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1. Triage & Rapid Emergency Assessment

Acute ischemic stroke is a time-critical neurological emergency ('Time is Brain'). The initial emergency department evaluation must follow the 10-25-45 rule:
? Door-to-physician assessment: Within 10 minutes.
? Door-to-non-contrast CT / CT angiography: Completed within 25 minutes.
? Door-to-CT interpretation: Completed within 45 minutes.
? Door-to-needle (IV thrombolysis): Target < 45 minutes (benchmark goal < 30 minutes).

2. Neurological Scoring & Clinical Examination

All candidates must be evaluated using the National Institutes of Health Stroke Scale (NIHSS):
? Score 1-4: Minor stroke.
? Score 5-15: Moderate stroke.
? Score 16-20: Moderate-to-severe stroke.
? Score 21-42: Severe stroke.

Mandatory Rapid Diagnostic Tests:
? Point-of-care capillary blood glucose (must rule out hypoglycemia < 60 mg/dL as stroke mimic).
? Coagulation panel (PT, INR, aPTT) and platelet count (thrombolysis is permissible prior to lab results unless coagulopathy or heparin/warfarin use is suspected).

3. Intravenous Thrombolysis Protocols (0 to 4.5 Hours)

Patients presenting within 4.5 hours of symptom onset (or last known normal) without contraindications are eligible for IV thrombolytic therapy.

Agent Selections & Dosing:
? Tenecteplase (TNK-tPA): Recommended first-line agent. Dose is 0.25 mg/kg IV single bolus over 5 seconds (maximum dose 25 mg). Does not require an infusion pump.
? Alteplase (rtPA): Alternative agent. Total dose 0.9 mg/kg (maximum 90 mg). Administer 10% of total dose as an IV bolus over 1 minute, followed by the remaining 90% infused over 60 minutes.

Absolute Contraindications to Thrombolysis:
? Intracranial hemorrhage on baseline CT scan.
? Significant head trauma or prior stroke within the past 3 months.
? Platelet count < 100,000/mm3 or INR > 1.7.
? Use of full-dose DOAC (Apixaban, Rivaroxaban) within the prior 48 hours with normal renal function.

4. Acute Blood Pressure Management Protocol

Strict blood pressure control is paramount to avoid hemorrhagic transformation:
? Prior to IV thrombolysis: Blood pressure must be lowered to Systolic < 185 mmHg and Diastolic < 110 mmHg.
? First-line antihypertensive agents:
  - Labetalol: 10 mg to 20 mg IV push over 1-2 minutes; may repeat or double every 10 minutes to maximum 300 mg.
  - Nicardipine IV infusion: 5 mg/hr, titrate up by 2.5 mg/hr every 5-15 minutes (maximum 15 mg/hr).
? During and after thrombolysis: Maintain BP < 180/105 mmHg for at least 24 hours post-thrombolytic administration.

5. Endovascular Mechanical Thrombectomy (EVT)

Mechanical thrombectomy with stent retriever or aspiration catheter is indicated for large vessel occlusions (LVO) of the internal carotid artery (ICA) or proximal middle cerebral artery (MCA M1 segment).

Eligibility Criteria:
? Within 0 to 6 hours of onset: Pre-stroke mRS 0-1, NIHSS >= 6, ASPECTS score >= 6 on non-contrast CT.
? Extended window (6 to 24 hours): Eligible if fulfilling DAWN or DEFUSE-3 trial perfusion mismatch criteria (target mismatch on CT perfusion or MRI diffusion-weighted imaging showing penumbra salvability).

6. Inpatient Subacute Care & Secondary Prevention

Post-thrombolysis Inpatient Directives:
? Neuro-checks and vital signs: Every 15 min for 2 hours, every 30 min for 6 hours, then hourly for 16 hours.
? Nil per os (NPO) until formal dysphagia screening is performed to prevent aspiration pneumonia.
? Repeat non-contrast CT at 24 hours post-thrombolysis before starting any antithrombotic medications.

Secondary Prevention Regimen:
? Antiplatelet therapy: Initiate Aspirin 160-325 mg daily starting 24 hours post-thrombolysis. For high-risk TIA (ABCD2 >= 4) or minor stroke (NIHSS <= 3), start dual antiplatelet therapy (DAPT: Aspirin 81 mg + Clopidogrel 75 mg daily) for 21 days.
? Lipid-lowering therapy: High-intensity statin therapy (Atorvastatin 80 mg daily or Rosuvastatin 40 mg daily) initiated regardless of baseline LDL level.